TREES (10.1)

Discrete Mathematics

TREE

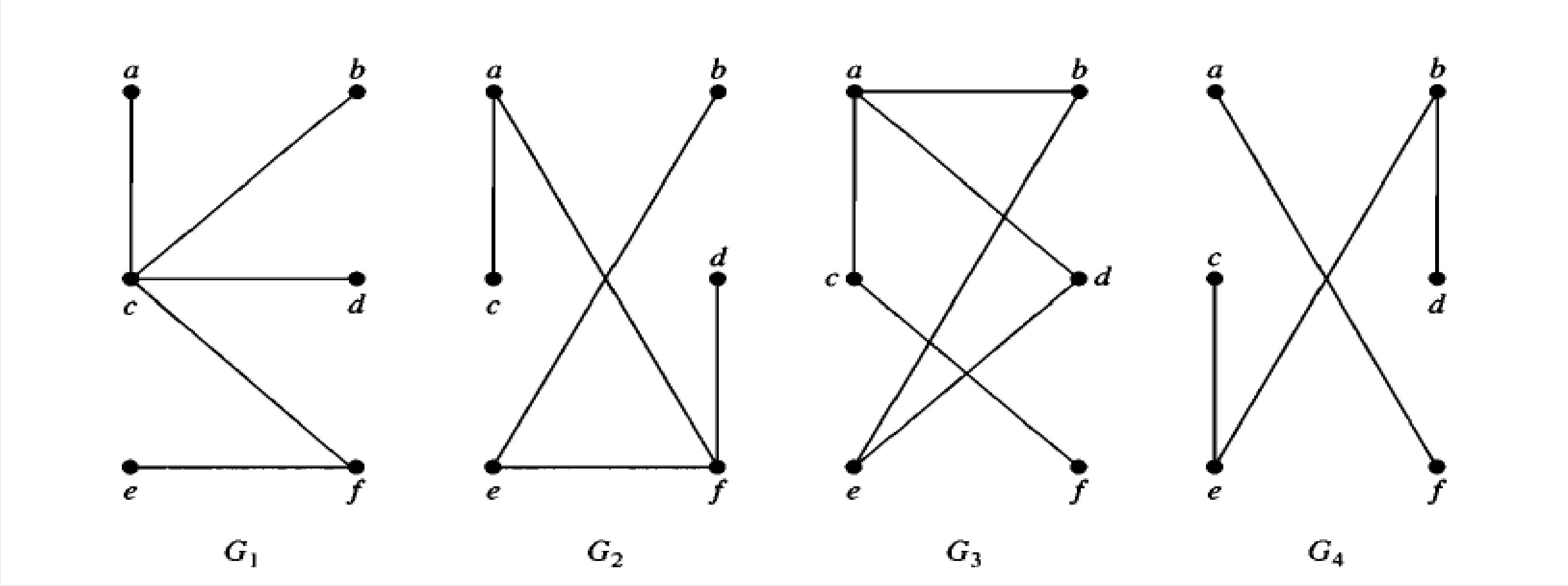
- DEFINITION : A *tree* is a connected undirected graph with no simple circuits.

➢ A tree cannot have a simple circuit.

➢ A tree cannot contain multiple edges or loops.

EXAMPLE

- Which of the graphs shown in following Figure are trees?



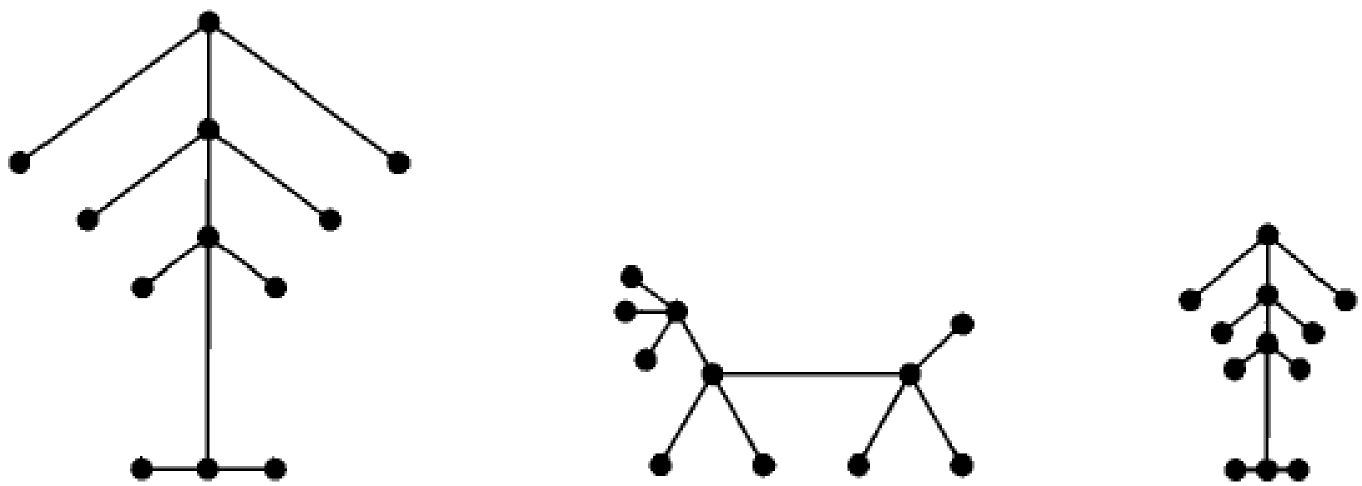
FOREST

- Any connected graph that contains no simple circuits is a tree.

- What about graphs containing no simple circuits that are not necessarily connected?
  - These graphs are called forests.
  - Each of their connected components is a tree.

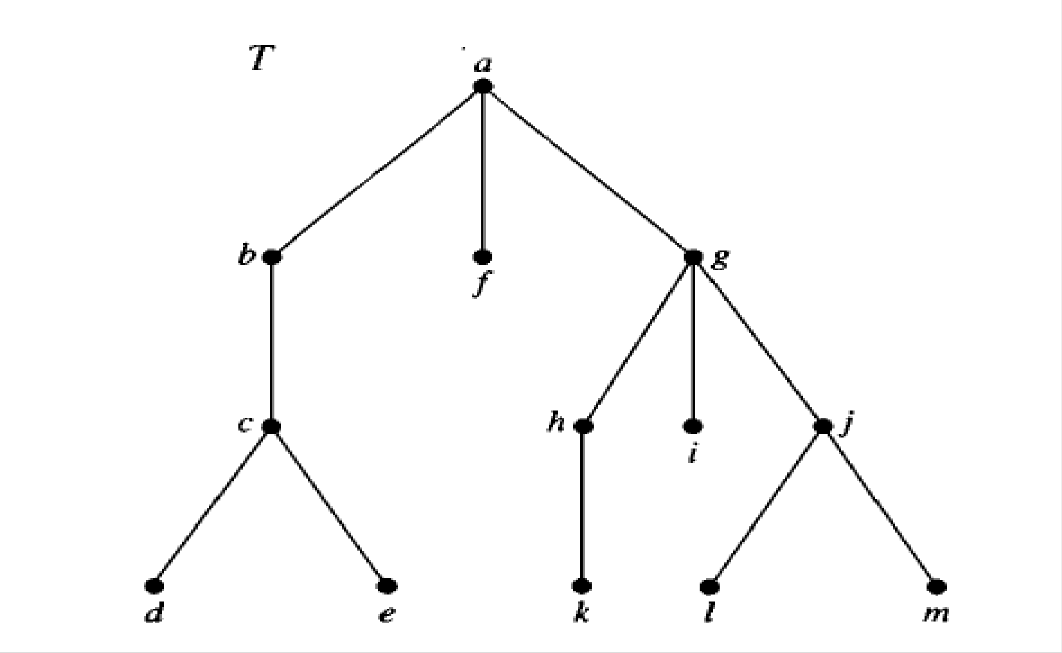
EXAMPLE

This is one graph with three connected components.



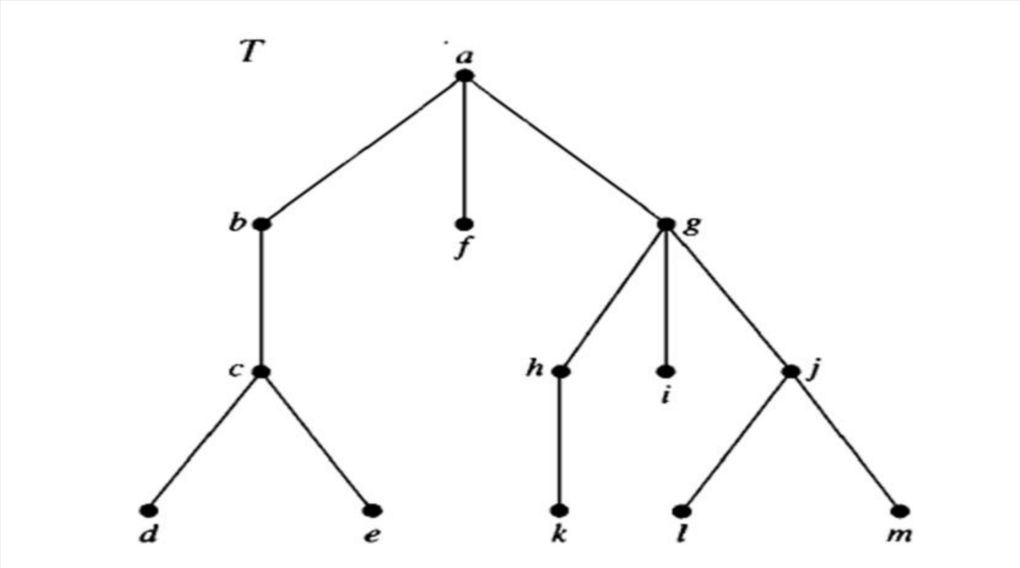
ROOTED TREE

- A *rooted tree* is a tree in which one vertex has been designated as the root and every edge is directed away from the root.



THE TERMINOLOGY OF TREES

- In the rooted tree *T* (with root *a*) shown in Figure , find the following vertices.



- The parent of c,
- The children of g,
- The siblings of *h*,
- All ancestors of e,
- All descendants of *b*,
- All internal vertices, and
- All leaves.
- What is the subtree rooted at g?

THE TERMINOLOGY OF TREES

- **Parent**

  Suppose that T is a rooted tree. If v is a vertex in T other than the root, the parent of v is the unique vertex u such that there is a directed edge from u to v.

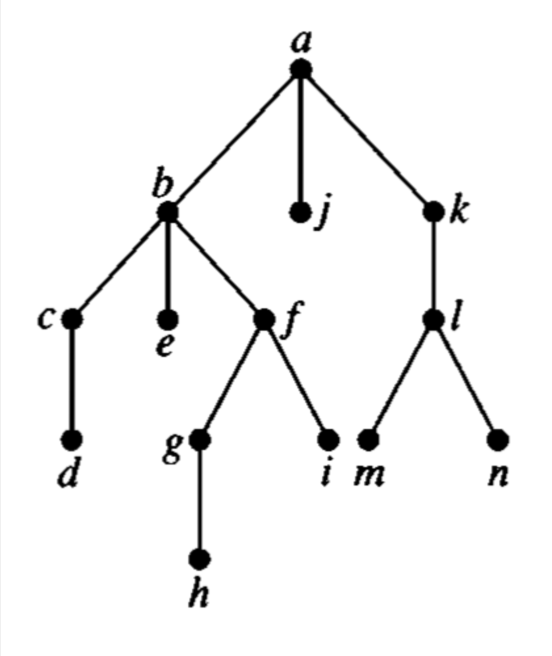
- **Child**
  If *u* is the parent of *v*, then *v* is called a *child* of *u*.
- **Siblings**
  Vertices with the same parent are called *siblings*.
- **Internal Vertices**
  Vertices that have children are called *internal vertices*.
  - The root is an internal vertex, unless it is the only node in the tree, in which case it is a leaf

THE TERMINOLOGY OF TREES

- **Ancestors**
  The *ancestors* of a vertex, other than the root, are the vertices in the path from the root to this vertex, excluding the vertex itself and including the root.
- **Descendants**
  The *descendants* of a vertex *v* are those vertices that have *v* as an ancestor.
- **Leaf**
  A vertex of a tree is called a *leaf* if it has no children.

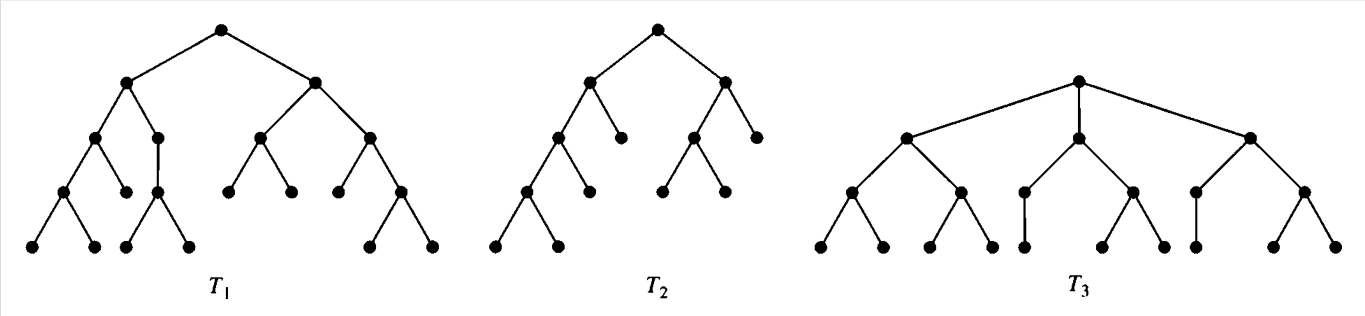
PROPERTIES OF TREES

HEIGHT OF TREE AND LEVEL OF NODES



- Node a is in level 0
- Nodes c, e, f, l are in level 2
- The tree height is 4, because the largest level of any vertex is 4
  - h is the node with max level

BALANCED TREE



- A tree is balanced if all of its leaves are in level h or $h - 1$ (h is the height of the tree)
- Which of these trees are balanced??

THEOREM 2

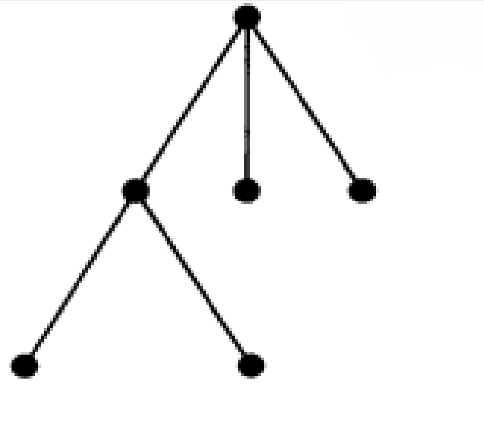
A tree with n vertices has $n - 1$ edges.

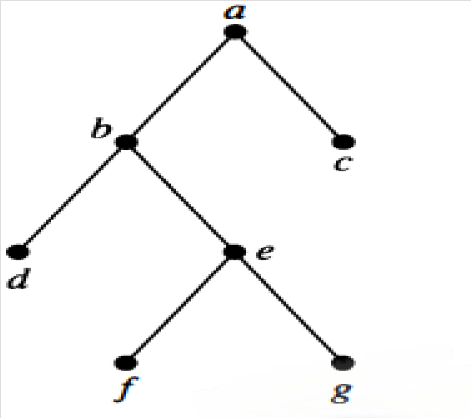
THEOREM 3

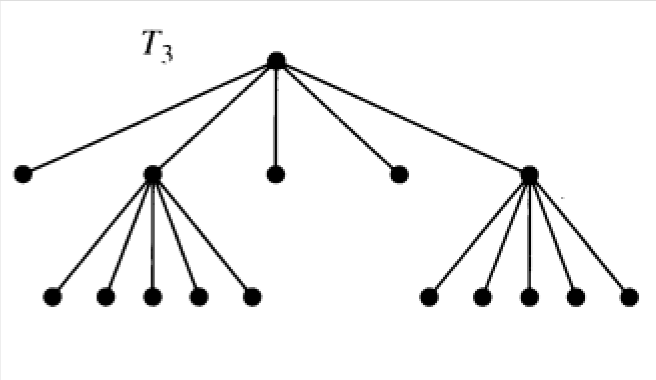
A full m-ary tree with i internal vertices contains a total of $n = mi + 1$ vertices

- ***m*-ary Tree**
  A rooted tree is called an *m-ary tree* if every internal vertex has no more than m children. The tree is called a *full m-ary tree* if every internal vertex has exactly m children.
- **Full m-ary tree:** an m-ary tree where every internal vertex has exactly m children

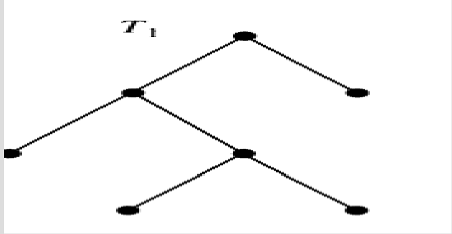
EXAMPLE







BINARY TREE
ORDERED ROOTED TREE



- **Binary Tree**
  An *m*-ary tree with $m = 2$ is called a *binary tree.*
- **Ordered Rooted Tree**
  An *ordered rooted tree* is a rooted tree where the children of each internal vertex are ordered.
- **Left and Right Child**
  In an ordered binary tree, the first child is called the *left child* and the second child is called the *right child*.
- **Left and Right Subtree**
  The tree rooted at the left child is called the *left subtree* and the tree rooted at the right child is called the *right subtree*.

THEOREM 4

A full m-ary tree with i internal vertices contains a total of $l = (m-1)i + 1$ leaves

EXAMPLE

Calculate the number of vertices in a full 5-ary tree with 35 internal vertices.

Also, find out the number of leaves.

EXAMPLE

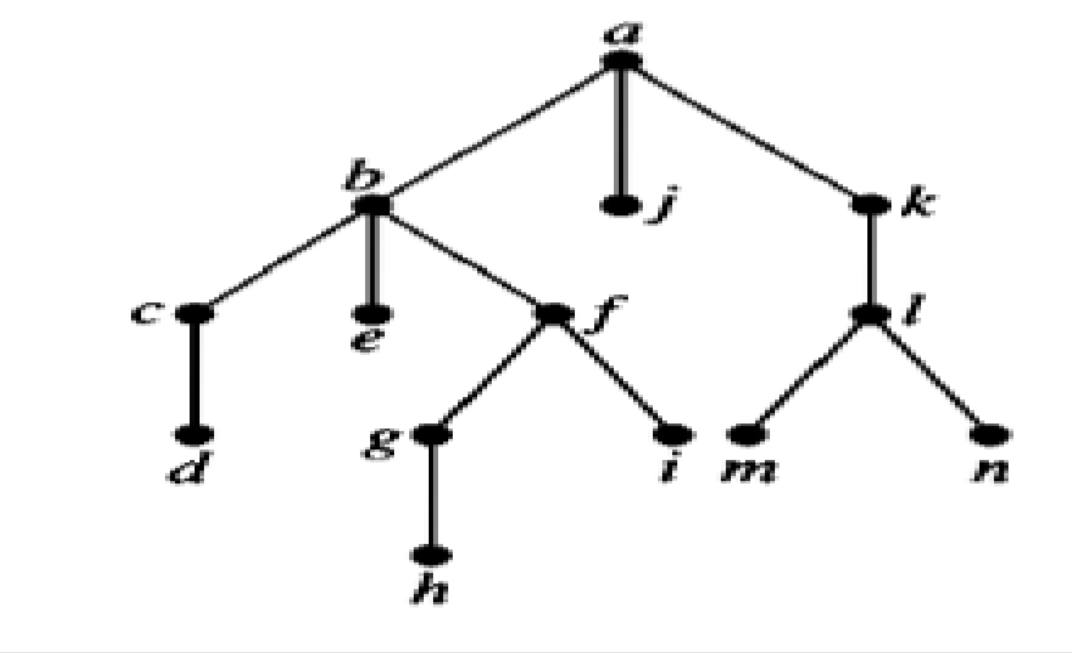
Calculate the number of vertices in a full 5-ary tree with 35 leaves.

Also, find out the number of internal vertices.

THEOREM 5

An m-ary tree of height h has at most m^h leaves

EXAMPLE



PRACTICE PROBLEM

- Exercise- 2,3,4,5,9,17,18,19,20